Daniel Guimarans

Current Position
Principal AI Scientist
Lyric
Barcelona, Spain (remote)

Additional Information
Email: daniel.guimarans@gmail.com
Website: dguimarans.net
Specialization: Algorithm Design, Heuristics, Optimization
Logistics & Transportation
Languages: English (IELTS 8.5), Spanish, Catalan

PROFESSIONAL SUMMARY

Principal AI Scientist with 20+ years of combined experience in academic and industry research in optimization and applied AI for logistics and transportation. PhD in Computer Science, with 40+ peer-reviewed publications and 3 completed PhD supervisions, built on extensive collaboration with industry and government partners. Delivered large-scale optimization systems for freight networks, port operations, and airline scheduling, with documented operational savings exceeding $300MM.

EDUCATION

Ph.D. in Computer Science, Cum Laude
Universitat Autònoma de Barcelona, Barcelona, Spain 2012
DISSERTATION - Hybrid Algorithms for Solving Routing Problems

M.Sc. in Industrial Computer Science & Advanced Production Techniques
Universitat Autònoma de Barcelona, Barcelona, Spain 2007
DISSERTATION - A Decision Support System for the Emergency Services Coordination Problem in a Road Accident
Sant Jordi Award Winner - Best Information Technology Master Thesis

Bachelor of Science, Physics
Universitat Autònoma de Barcelona, Barcelona, Spain 2005

Minor, Applied Mathematics
Universitat Autònoma de Barcelona, Barcelona, Spain 2004

PROFESSIONAL EXPERIENCE

Lyric February 2026 – Present
Barcelona, Spain (remote) Principal AI Scientist

- Technical lead driving the heuristics strategy for multiple initiatives within Lyric's decision products, including both initiatives below
- Defined and delivered the heuristics-based solution for supply planning, turning the founders' MVP vision into a solution now powering large-scale problem solving in the Lyric platform, prioritizing solving speed and plan explainability
- Architected and built (in Rust) a new heuristics-based solver covering multiple routing flavors, closing in on state-of-the-art solution quality at a fraction of the computational time and scaling to problem sizes unreachable by commercial solvers (tested on instances of several tens of thousands of customers with multiple side constraints)
- Partnering with NVIDIA to develop and test the capabilities of their GPU-based cuOpt solver on large-scale routing problems

- Development of an agentic framework for self-development and self-validation of customer-specific optimization solutions, from problem description to a fully functional, deployable solution; subject of a U.S. provisional patent filed in August 2026

Amazon February 2020 – January 2026
Barcelona, Spain Principal Research Scientist (L7)

- Lead optimization initiatives for Amazon's global transportation and logistics network delivering operational efficiencies and cost savings exceeding $100MM annually across multiple business units
- Develop scalable algorithms for end-to-end network optimization supporting millions of daily shipments across global fulfillment operations
- Develop strategic research roadmaps and system architectures enabling new business line expansion while optimizing resource allocation across established middle-mile and last-mile operations
- Design optimization frameworks supporting network efficiency improvements and capacity planning across multiple geographic regions and service tiers
- Previous Role: Senior Research Scientist (L6) – February 2020 - April 2022; Luxembourg, Luxembourg

Monash University February 2019 – January 2020
Melbourne, Australia Research Fellow - Optimization & AI

- **Melbourne Metro Optimization:** Delivered congestion-minimization system using large-scale Public Transport Victoria datasets, directly impacting daily commuter experience
- **Digital Twin Development:** Created open-source discrete-event simulation engine for Queensland's coal rail system and ground operations at European airports
- Developed hybrid simulation-optimization techniques for complex stochastic problems across aviation, logistics, and transportation
- Advanced state-of-the-art approaches in stochastic optimization with direct industrial applications

Amsterdam University of Applied Sciences October 2015 – January 2019
Amsterdam, Netherlands Senior Researcher - Lecturer

- Led research in aviation operations optimization and disruption management while establishing new academic programs
- **Industry Partnerships:** Delivered optimization solutions for Amsterdam's green taxi fleet in collaboration with Amsterdam City Council and supervised students hosted by KLM and Schiphol Airport
- **Program Leadership:** Developed and launched Aviation Logistics curriculum and Aviation Operations master program, coordinating a team of multi-disciplinary researchers and lecturers
- Taught advanced courses in data analysis, optimization, and operational resilience to 200+ students annually

NICTA (National ICT Australia) March 2013 – October 2015
Sydney, Australia Researcher

- **Port Botany Optimization (2013-2014):** Maximized rail capacity at Australia's busiest container port through simulation and scheduling algorithms, collaborating with Sydney Ports Corporation and NSW Ports. Project won Best Industry Project Award and generated significant media coverage
- **Virgin Australia Crew Optimization (2014):** Analyzed and optimized crew productivity and rostering practices for E190 fleet operations
- **Canberra Light Rail Impact Analysis (2015):** Minimized traffic disruption from new light rail system through predictive simulation modeling for ACT Government
- **Fleet Logistics Solutions:** Developed routing optimization systems for multiple commercial clients

Universitat Autònoma de Barcelona September 2005 – February 2013
Barcelona, Spain PhD Researcher & Postdoctoral Researcher

- **Postdoctoral Research (2012-2013):** Advanced hybrid optimization methodologies combining constraint programming, metaheuristics, and simulation for transportation applications

- **PhD Research & Teaching (2005-2012):** Pioneered hybrid algorithms for vehicle routing and transportation optimization problems
- **Emergency Services Optimization:** Developed distributed decision support systems for emergency services coordination
- **Software Commercialization:** Created and licensed two optimization platforms (ITSLogisim Optimization Suite and Simulation Suite) to Digital Aeronautics Engineering Services
- Supervised 20+ graduation theses while maintaining active research output

SELECTED INDUSTRY-SPONSORED PROJECTS

- **Peak-hour Passenger Shifting (2019)** - Monash University & Public Transport Victoria — Congestion reduction system for Melbourne's rail network affecting 500K+ daily commuters
- **Green Taxi Fleet Optimization (2018)** - Amsterdam University & Amsterdam City Council — Data analysis and simulation optimization of taxi charging ranks for sustainable urban transportation
- **Light Rail Traffic Impact (2015)** - NICTA & ACT Government — Minimization of traffic disruption for Canberra's new light rail system
- **Virgin Australia Crew Optimization (2014)** - NICTA & Virgin Australia — Productivity analysis and rostering optimization for E190 fleet operations
- **Port Botany Rail Optimization (2013-2014)** - NICTA & NSW Ports — Container freight train operations simulation and capacity maximization — Impact: $AUD 210MM+

INTELLECTUAL PROPERTY

- Automated System for Producing and Validating Mathematical Optimization Software Packages - **U.S. Provisional Patent Application No.: 64/103,812**; Filed: August 2026 by ChainBrain Inc. (dba Lyric) - Inventors: Faheem Faheem, Ugo Rosolia, Abhishek Nahar, Amit Kumar, Ratnaji Vanga, Daniel Guimarans, Pradyot Mohanty, Nandini Gaur
- ITSLogisim Simulation and Optimization Suite 1.0 (2010): Distributed simulation and optimization plaform for solving vehicle routing problems – Licensed to Digital Aeronautics Engineering Services

KEY ACHIEVEMENTS & RECOGNITION

- **Best Industry Project Award** (NICTA 2014) - Port Botany Capacity Assessment
- **Sant Jordi Award** (2008) - Best Information Technology Master Thesis, Catalan Government
- **Media Recognition** - Featured in Sydney Morning Herald, The Australian, Lloyd's List for groundbreaking port optimization work
- **40+ Peer-Reviewed Publications** in top Operations Research and Transportation journals and international conferences
- **Software Licensing** - Two commercial optimization platforms successfully licensed to industry
- **Patent (Provisional)** - Co-inventor, U.S. provisional patent application for automated optimization software generation and validation (filed 2026)
- **PhD Supervision** - 3 successful PhD graduates in optimization and logistics (Edinburgh, Open University of Catalonia, Universitat Autònoma de Barcelona)
- **International Recognition** - Keynote and invites presentations at major conferences across 3 continents

SELECTED INDUSTRY-RELEVANT PUBLICATIONS

- "Constraint-based Robust Planning and Scheduling of Airport Apron Operations through Simheuristics," *Annals of Operations Research*, 2023
- "A Stochastic Approach for Planning Airport Ground Support Resources," *International Transactions in Operational Research*, 2022
- "A Simheuristic Algorithm for the Capacitated Location Routing Problem with Stochastic Demands," *Journal of Simulation*, 2021
- "Peak-hour rail demand shifting with discrete optimisation," *Principles and Practice of Constraint Programming – CP*, 2019
- "A Bi-objective Approach for Scheduling Ground Handling Vehicles in Airports," *Computers & Operations Research*, 2016
- "Rich Vehicle Routing Problem: Survey," *ACM Computing Surveys*, 2014

Note: Complete list of 40+ peer-reviewed publications available at dguimarans.net

PROFESSIONAL ACTIVITIES

Editorial & Review Activities: Journal Reviewer for European Journal of Operations Research, Computers & Operations Research, Annals of Operations Research

Conference Activities: Program Committee Member for IJCAI, ICAPS, EURO, Winter Simulation Conference

Industry Partnerships: Government Agencies (Public Transport Victoria, ACT Government, NSW Ports, Amsterdam City Council), Airlines (Virgin Australia, KLM), Technology (Amazon), Ports & Infrastructure (Port Botany, European airports)